BUILD YOUR IDEA THROUGH AI

From a notebook to a Telegram bot and an Android app

How one fitness-coach idea became three surfaces on one backend, and what the AI actually did.

Notebook
rough notes

→

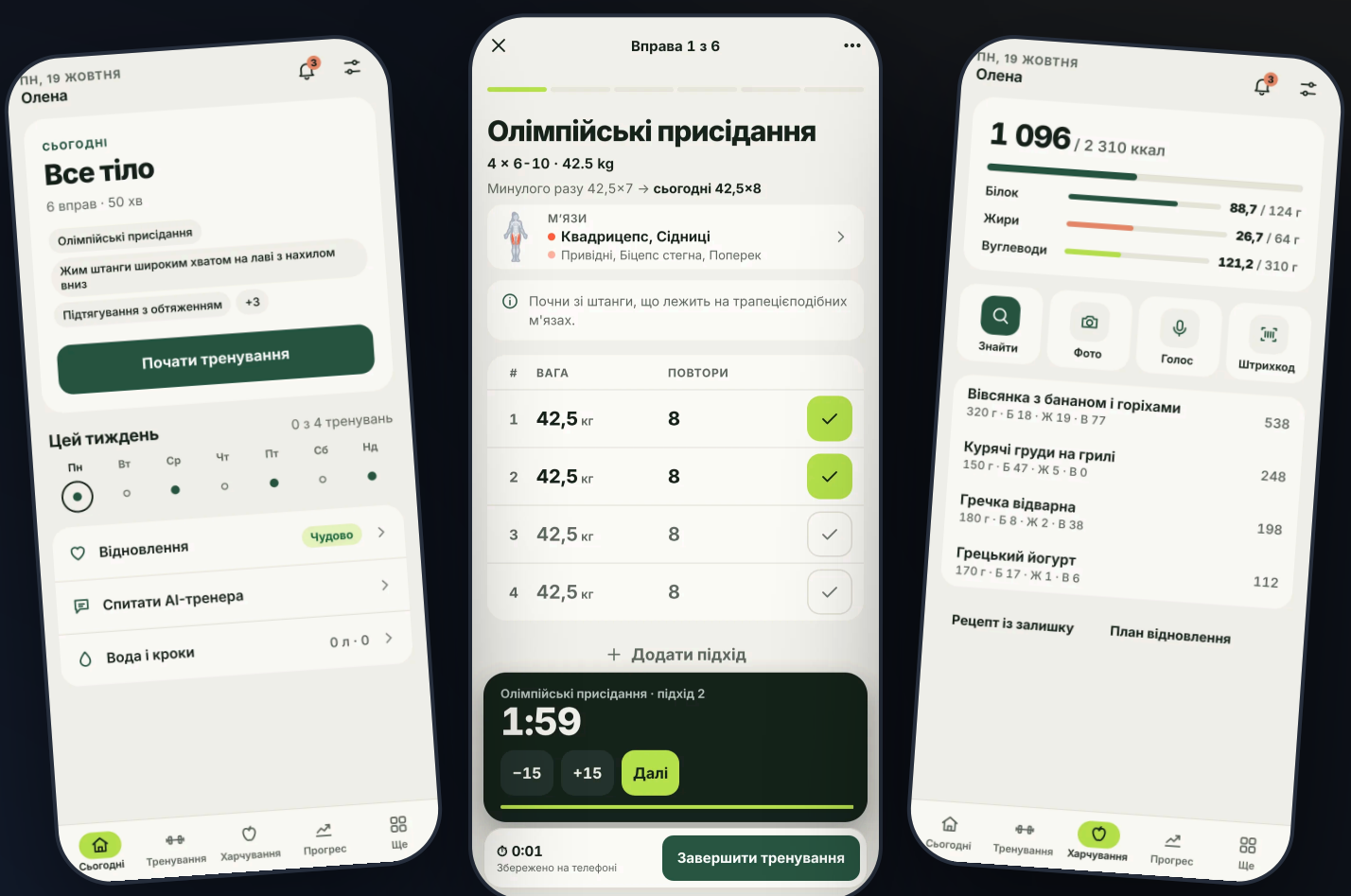
Bot
Telegram chat

→

Mini App
logger, food, plan

→

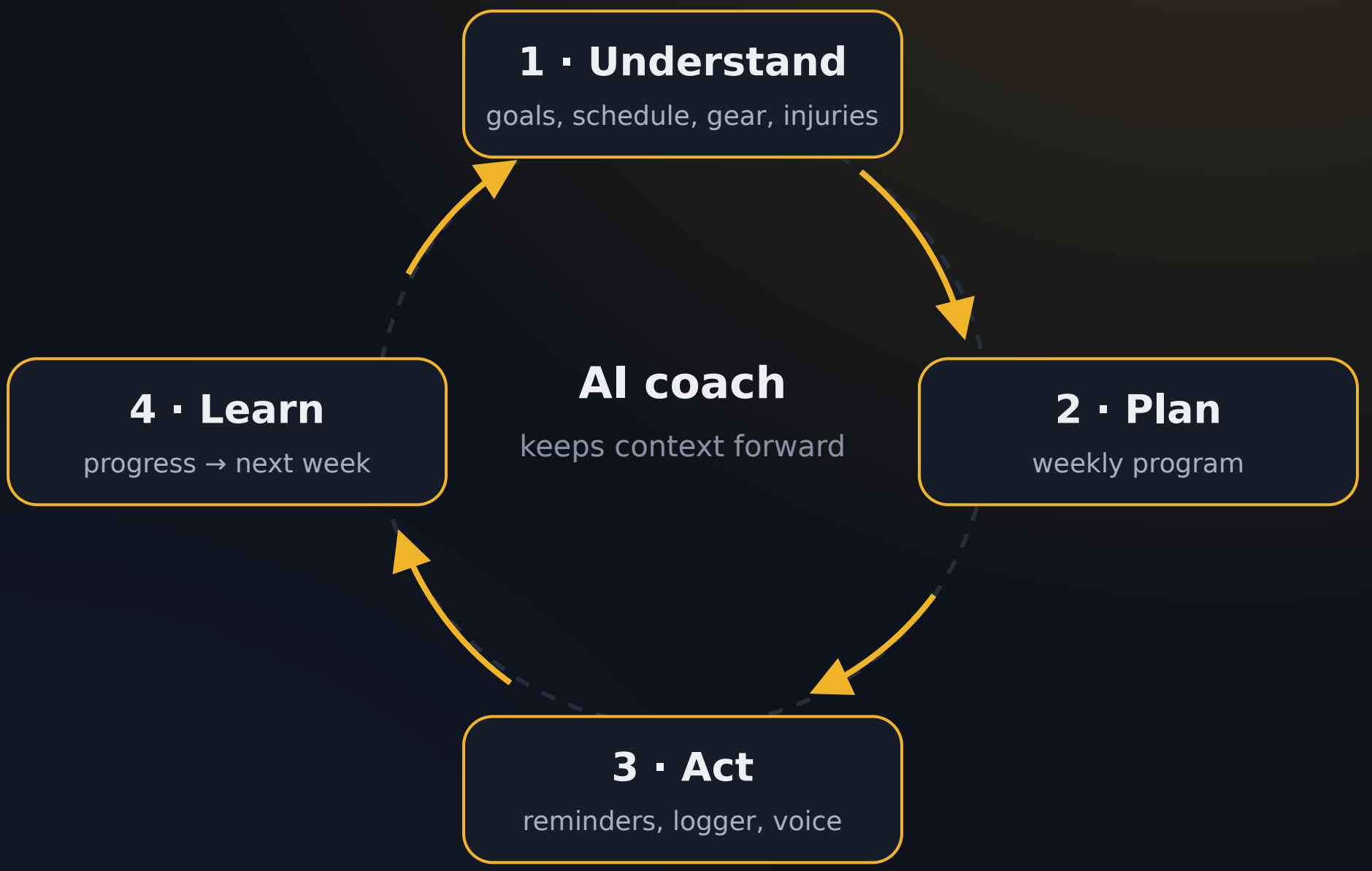
Android
Kotlin MP + Compose



STEP 1 · ONE COMPLETE LOOP

Don't start with a platform. Start with a loop.

Telegram removed all setup friction: no install, no sign-up. A short interview is enough to start coaching.



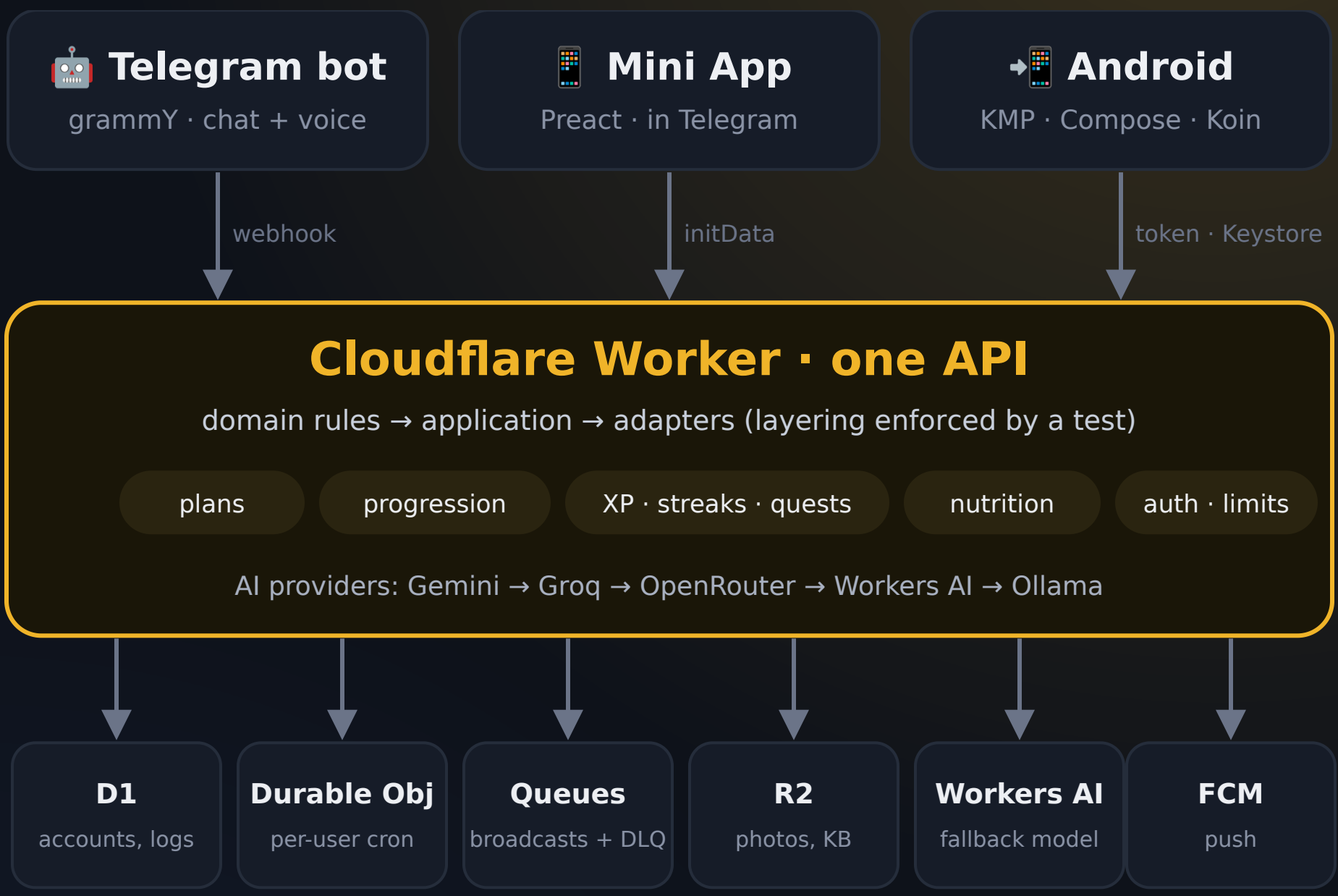
0
installs to try it

1
chat to onboard

uk · en
languages from day one

STEP 3 · MOBILE WITHOUT A SECOND PRODUCT

Three surfaces. One backend.



Why it matters

A workout logged on Android shows up in Telegram instantly.

Progression rules live in one place, not three.

Each surface only owns its UI: chat, web view, native.

STEP 4 · AI AS AN ACCELERATOR

The model call is one box in the pipeline

👤 **User context**
goals, equipment, injuries, history, language

↓

🧩 **Prompt + schema**
structured JSON contract, not free text

↓

🔁 **Provider fallback**
Gemini → Groq → OpenRouter → Workers AI → Ollama, with budget and retries

↓

🛡️ **Validate against the user**
no barbell for a home user, swap exercises that hit an injury, sane sets/reps

↓

📋 **Deterministic fallback**
a catalog plan if every model fails: the user never sees an error

↓

💾 **Save + coach**
D1 state, reminders, follow-ups, progress

STEP 5 · TEST THE WHOLE JOURNEY

Proof, not vibes

1,457
backend tests (Node)

92
Workers-runtime tests

250+
Android host tests

100+
D1 schema migrations

10
architecture decision records

5
AI providers behind one interface

CI on every PR: typecheck, lint, CodeQL, architecture rules, Android lint + build. Secrets and signing keys never touch the repo.

…and a real tester still finds what tests miss

· History list didn't scroll on a small phone
· Keyboard covered the field you were typing in
· Rest timer forgot that 90 s was your choice, not 60 s

THE LESSON

Pick the lowest-friction place to test one user problem.

Expand only when the next surface improves the same loop.

📓 Notebook → write the loop in one sentence

🤖 Bot → test it with real people this week

📱 Mini App → when chat gets too cramped

📲 Native → when you need sensors, push, offline

Open source → github.com/sanych44474/trix